

Nanodiamond Reinforced Poly(dimethylsiloxane)-Based Polyurea with Self-Healing Ability for Fouling Release Coating

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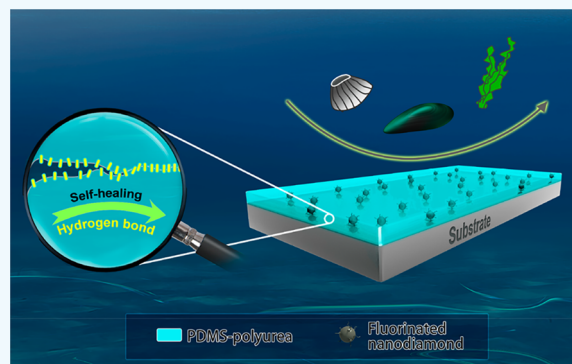
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Supporting Information

ABSTRACT: Silicone-based fouling-release coatings (FRCs) have received increasing interest because they are environmentally friendly. However, the coatings with poor mechanical properties and low adhesion to substrate are difficult to repair after damage, which limits their applications. In this study, poly(dimethylsiloxane)-based polyurea (PDMS-PUa) was modified by adding nanodiamond with perfluoroalkyl attached on the surface. The modified PDMS-PUa coating has high water contact angle (124°) and low surface energy (18 mJ m^{-2}) due to the presence of the self-enriched perfluoro groups and polysiloxane on the surface. The introduction of fluorinated nanodiamond improves the tensile strength of PDMS-PUa from 1.2 to 1.6 MPa and the adhesion strength from 1.5 to 2.1 MPa. Particularly, the modified PDMS-PUa exhibits self-healing performance at room temperature because of the hydrogen bonding between urea groups. Marine field tests show that the coating can keep fouling-release ability after immersion in natural seawater for four months.

KEYWORDS: biofouling, silicone, nanodiamond, fluoropolymer, self-healing



INTRODUCTION

With the development of marine industries, more attention is paid to the impact of marine biofouling. The accumulation of unwanted marine organisms on underwater surfaces forms marine biofouling. It increases the drag force and fuel consumption of vessels, accelerates corrosion, clogs seawater pipelines, and facilitates species invasion to different marine areas.^{1–4} To protect marine ecology, environmentally friendly and efficient antifouling technologies are required. Recently, some progress has been made in such technologies. For examples, fouling-release coatings (FRCs),^{5–13} degradable polymers with dynamic surfaces,^{15,16} biomimetic materials,^{17–22} the ultraviolet irradiation method,²³ and ultrasound treatment^{24,25} have been developed. FRCs based on silicone have received increasingly interest because they are biocide-free and have hydrodynamic drag-reducing performance.

Silicone-based FRCs promote the removal of accumulated organisms by shear force because of their weak adhesion to organisms. This fouling-release ability arises from the low elastic modulus, low surface energy, and low surface roughness. In particular, the elasticity allows the coating to form a physically dynamic surface where fouling organisms weakly attach.² Unfortunately, these characteristics also make FRCs poorly adherent to substrates and easily damaged by scratching and puncturing.⁶ Because the polysiloxane chains are covalently cross-linked, the mechanically damaged coating

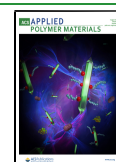
cannot be repaired. These poor mechanical properties limit their application.

To improve the mechanical performance of FRCs, urethane and epoxy moieties have been used to modify silicone. Webster et al.²⁶ developed silicone–polyurethane coatings with toughness and good adhesion from the polyurethane. Fang et al.²⁷ prepared silicone–polyurea with improved tensile strength ($\sim 3 \text{ MPa}$) and storage modulus ($\sim 10 \text{ MPa}$). Siripiprom et al.²⁸ prepared a UV-curable cycloaliphatic epoxy–siloxane FRC, which exhibited good impact resistance and surface stability. However, these materials cannot repair themselves when mechanical damage occurs. It was reported that poly-(dimethylsiloxane)-based polyurea (PDMS-PUa) exhibited remarkable self-healing performance.^{29,30} When the coating is cut, it is capable of repairing itself within 2 days at room temperature. The hydrogen bonds between the ureas in different chains are responsible for the self-healing. Note that this optimized self-healing ability is achieved at a low hard segment content (10 wt %), so the PDMS-PUa is not mechanically robust enough. Actually, self-healing and

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mechanical strength appear to be contradictory properties. Silicone coatings modified by urethane, urea, or epoxy generally cannot outperform traditional silicone elastomer in terms of fouling release since the elastic modulus increases.

It is well-known that incorporating nanofillers in a polymer matrix can effectively improve mechanical strength. For this purpose, carbon nanotube, graphene oxide, and natural sepiolite have been added to silicone elastomer.^{31–33} Nanodiamond has extremely high hardness and superior abrasive resistance, as diamond has the highest hardness on earth.³⁴ Nanodiamond also has abundant oxygenic functional groups on surface, which makes surface chemical modification possible. In this study, we prepared modified nanodiamond particles by grafting perfluoro groups on their surfaces and introduced them to PDMS-PUa to form FRC. The fluorinated nanodiamond not only improves the mechanical properties of PDMS-PUa but also lowers the surface energy, which further improves the fouling release ability. The modified PDMS-PUa coating is biocide-free and hence eco-friendly. The surface wettability, roughness, and mechanical properties of the coating were investigated. In particular, the antifouling efficacy in natural seawater was studied by a marine field test. Our aim is to develop an environmentally friendly fouling-release system with improved mechanical properties.

■ EXPERIMENTAL SECTION

Materials. α,ω -Aminopropyl-terminated poly(dimethylsiloxane) ($M_n = 2000 \text{ g mol}^{-1}$) was purchased from Hangzhou Silong Mater. Tech. Co., Ltd. Tetrahydrofuran (THF, $\geq 99\%$) from Sinopharm Chemical Reagent Co., Ltd., was purified by distillation with CaH_2 prior to use. Toluene (99.5%, Guangzhou Chemical Reagent Factory) was purified by refluxing with sodium for 12 h before distillation. Concentrated hydrochloric acid (36%–38%), nitric acid (65%–68%), and sodium hydroxide from Guangzhou Chemical Reagent were used as received. Isophorone diisocyanate (IPDI, $>99\%$) and lithium aluminum hydride (LiAlH_4) solution (2.5 mol L^{-1} solution in THF) from Aladdin were used as received. Nanodiamond (ND) was kindly presented by Wei Xing Chemical Co., Ltd., and dried at 130°C under vacuum for 48 h prior to use. 1H, 1H, 2H, 2H-perfluorodecyltriethoxysilane (FAS17, $\geq 97\%$) from Adamas was diluted with anhydrous toluene prior to use. PDMS (26000 g mol^{-1}) and its curing agent poly(diethoxysiloxane) for preparing elastomer were purchased from Gelest and used as received. Artificial seawater (ASW) was prepared according to ASTM D1141-98 (2003). All other reagents and solvents were analytical grade and used without further purification.

Preparation of Fluorinated Nanodiamond (FND). The FND was prepared in three steps: oxidation, reduction, and silanization.^{35,36} The nanodiamond produced by the explosive method has hydroxyl, carboxylic, carbonyl, and methylene groups are generated on surface. The oxidation was applied to convert most groups into hydroxyl and carboxylic groups. Afterward, the carboxylic groups were reduced to hydroxyl groups, so that abundant hydroxyl groups are present for silanization.

Oxidation: 1.0 g of ND was dispersed in 30 mL of hydrochloric acid and nitric acid mixture ($v/v = 3/1$). The mixture was stirred at room temperature for 3 days, diluted with 120 mL of deionized water, separated by centrifugation at 8000 rpm, and rinsed with deionized water. Then, the ND was added in 50 mL of 0.1 mol L^{-1} NaOH aqueous solution and heated at 90°C for 2 h. The mixture was also diluted, centrifuged, and rinsed as stated above. Next, the ND was added in 50 mL of 0.1 mol L^{-1} hydrochloric acid solution and heated at 90°C for 2 h. The reaction mixture was washed until the washing solution became weakly acidic. The oxidized ND was obtained by centrifugation and dried at 50°C under vacuum for 24 h.

Reduction: 0.5 g of oxidized ND was dispersed in 30 mL of anhydrous THF in an ice–water bath under ultrasonication. The

suspension of ND was bubbled with nitrogen, and 8.5 mL of LiAlH_4 solution was then added to the system under nitrogen. The reactive mixture was refluxed for 24 h and then with deionized water, centrifuged at 8000 rpm for 10 min, and rinsed with deionized water. Next, the ND was treated with 100 mL of 6.0 mol L^{-1} NaOH solution for 24 h, centrifuged at 8000 rpm for 10 min, and subsequently with 100 mL of 0.1 mol L^{-1} hydrochloric acid solution for 2 h. The mixture was diluted, centrifuged, rinsed, and dried at 50°C for 24 h under vacuum, yielding the reduced ND with enriched hydroxyl on the surface.

Silanization: 0.3 g of the reduced ND was dispersed in 40 mL of toluene in an ice–water bath under ultrasonication. Next, 40 mL of 0.01 mol L^{-1} FAS17 toluene solution was added under nitrogen, and the reactive mixture was stirred at room temperature overnight. The mixture was then centrifuged, and the precipitate was extracted in a Soxhlet extractor for 24 h with methanol as a solvent to remove the unreacted or physically adsorbed fluorosilane molecules. Finally, the product was dried at 50°C under vacuum, yielding the fluorinated ND (FND).

Preparation of PDMS-PUa Coating Modified with Nanoparticle. PDMS-PUa was synthesized following previous report.²⁹ In a 250 mL three-necked round-bottom flask equipped with a dropping funnel, a condenser, and a N_2 inlet, 4.46 g of isophorone diisocyanate dissolved in 25 mL of THF was introduced. 40.0 g of aminopropyl-terminated dissolved in 100 mL of THF was added dropwise into the flask under N_2 in 1 h. The mixture was allowed to react for 3 h at 25°C . The product was then precipitated in excess water and dried under vacuum for 24 h. The structure of PDMS-PUa was confirmed by ^1H NMR (Bruker AV600 NMR, 600 MHz) with CDCl_3 as solvent (Figure S1). The polymer has a number-average molar mass (M_n) of 12100 g mol^{-1} and a polydispersity (D_M) of 2.9, determined by Agilent gel permeation chromatography with THF as solvent (1260 Infinity with two PLgel columns).

Unmodified nanodiamond (ND) or fluorinated nanodiamond (FND) was uniformly dispersed into a PDMS-PUa matrix by solution blending, which improves the dispersion effect of nanoparticles in bulk polymer.³⁷ ND or FND was dispersed in THF by ultrasonication for 30 min under an ice–water bath to prepare a 0.15 mg mL^{-1} nanoparticle suspension. PDMS-PUa was dissolved in dried THF in another vial to prepare a 0.20 g mL^{-1} polymer solution. Next, the nanoparticle suspension was added dropwise into PDMS-PUa solution under vigorous stirring. The resulting mixture was heated at 50°C to remove some solvent until the viscosity reached $\sim 500 \text{ mPa}\cdot\text{s}$.

The composite coating was prepared by the solution casting method. The polymer solution was applied on the substrates by a pipet, and the solution was dried at 25°C for 3 days to yield the coating. The substrate is epoxy panel consisting of electrical glass cloth and epoxy resin (Guangzhou Jingke Chemical Consumables) or glass slide (Shanghai Sangon Biotech). The ND/FND content in the dry coating was 0.1 wt %. The coating with ND was designated as PDMS-PUa-ND for convenience, and the one containing FND was designated as PDMS-PUa-FND. The thickness of the dry coating was $\sim 200 \mu\text{m}$ for all the coatings, except for the free-standing film used for the tensile test ($\sim 500 \mu\text{m}$). The standard deviation was $20 \mu\text{m}$ across different composite coatings and was $10 \mu\text{m}$ across multiple batches for each composite. PDMS-PUa containing 0.05 and 0.2 wt % of ND/FND as well as 0.1 wt % of nanosepiolite and halloysite nanotube has also been prepared by the same procedure. In addition, a polyurea coating without nanodiamond and a standard PDMS elastomer coating (PDMS/curing agent, wt/wt = 10:1) were prepared through the same solution casting method.

Characterization. Raman Spectroscopy. Raman spectroscopy measurement of FND was performed on a Raman microscope system (LabRAM ARAMIS) with an Nd:YAG laser (532 nm) and an actual output power of 148 mW. The spectroscopy was collected at 32 scans, and the resolution was 4 cm^{-1} .

Transmission Electron Microscope (TEM). TEM images of ND and FND were collected by a Hitachi (Japan) H-7650 TEM with an accelerating voltage of 20 kV. The nanoparticles were dispersed into

Scheme 1. Preparation of FND-Modified PDMS-PUa Coating

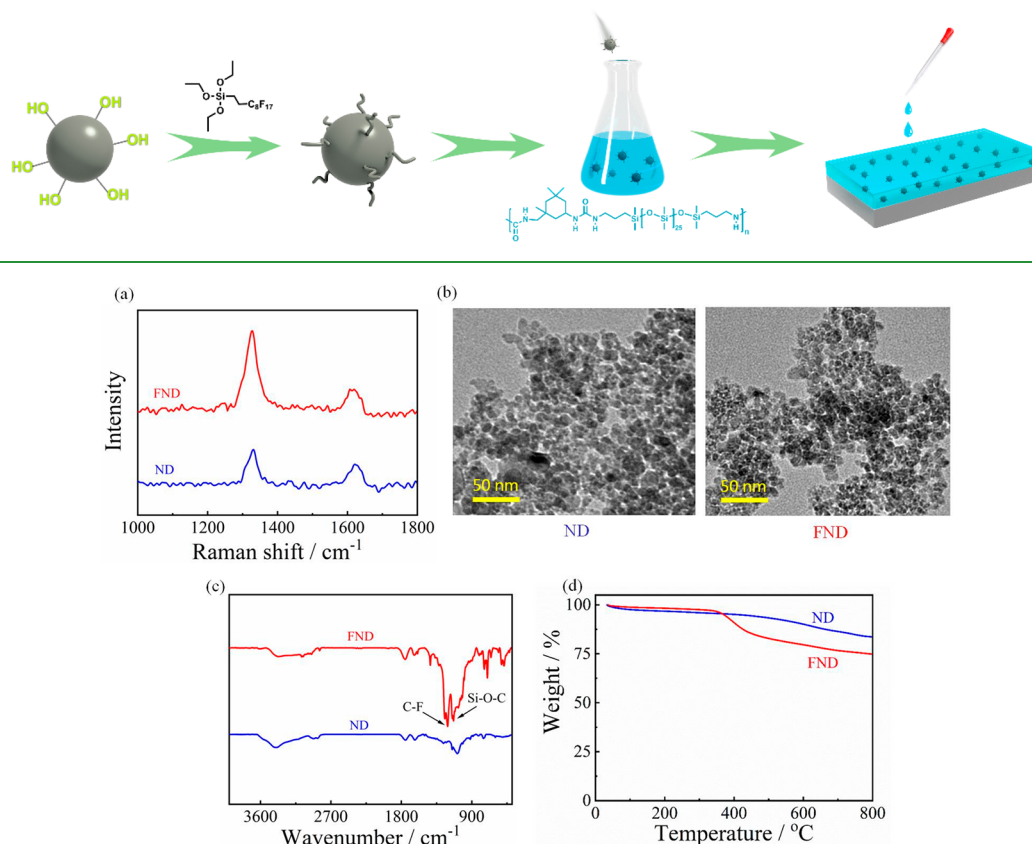


Figure 1. Raman spectra (a), TEM images (b), FTIR spectra (c), and TGA curve (d) of ND and FND.

ethanol under ultrasonication, and then three drops of the suspension were added to carbon-coated copper grid. The test was conducted after evaporation of ethanol.

Scanning Electron Microscope (SEM). SEM images of nanoparticles and the composite coatings were collected by a Quanta 200 scanning electron microscope (SEM, Philips-FEI Corporation, Netherlands) at 10.00 kV. Before the observation, the samples were sputter-coated with gold to minimize sample charging.

Fourier Transform Infrared Spectroscopy (FTIR). The FTIR spectra were recorded on an IR spectrometer in the range of 4000–500 cm^{-1} (Bruker VECTOR-70). The spectra were collected at 64 scans by using the KBr disk method, and the spectral resolution was 4 cm^{-1} .

Thermal Gravimetric Analysis (TGA). Thermal stability of the nanodiamond was studied by TG 209F1 thermal analysis equipment (NETZSCH Instruments). The test was conducted from 30 to 600 $^{\circ}\text{C}$, and the heating rate was 10 $^{\circ}\text{C min}^{-1}$ with nitrogen gas (50 mL min^{-1}).

Adhesion Test. The adhesion of coatings to substrates was characterized by using the pull-off test according to ASTM D4541. The adhesion of the coating to the epoxy resin substrate was measured by a DeFelsko PosiTest adhesion tester. By use of epoxy adhesive, an aluminum dolly (diameter: 20 mm) was adhered to the PDMS-based coatings. The dolly was detached from the coating (speed: 0.2 MPa s^{-1}), and the data were recorded by the tester. Each coating was tested for five times to get an average value.

Self-Healing Test. The modified PDMS-PUa was cut by a scalpel, gently jointed together, and allowed to self-repair at 25 $^{\circ}\text{C}$ for different amounts of time, under ASW or in air. Next, the composite was photographed or observed with an optical microscope. A tensile test was also performed to quantitatively study the efficiency of self-healing process.

Tensile Test. Dog-bone-shaped specimens were obtained by a sheet punching machine. The tensile test was conducted on a GOTECH AI-7000S controller equipped with a 100 N load cell (speed: 25 mm min^{-1}), following ASTM D412. Five specimens of each material were tested to get an average value.

Static Water Contact Angle (WCA) and Surface Energy (SE) Measurements. A DataPhysics OCA40 contact angle system was used to measure the static contact angle of deionized water and diiodomethane. The volume of sessile droplet was 3 μL . Before testing, the surface of each polymer coating was cleaned with dry nitrogen. Five different regions of each coating were measured to get an average value. The Owens–Wendt–Rabel–Kaelble model was used to determine SE.

Surface Roughness and Topography Measurements. The surface roughness and topography of the samples were characterized with a BMT Expert 3D surface profile measurement system. Five measurements, with a line scan of 2 mm in length, were performed on different regions of each sample, and the average value (R_a) was reported. Surface topography of an area of 1 mm $\times$ 1 mm was determined with the area-scan technology.

Pseudobarnacle Adhesion Test. The test was conducted following ASTM D5618 (2011). An aluminous cylindrical stud (diameter: 10 mm; height: 10 mm) was glued to the coating with epoxy adhesive (Araldite Rapid Epoxy Adhesive, Huntsman), and the epoxy adhesive was cured for 3 days (16 h for complete cure according to the instructions). Then, a SUNDOO SH-500 force gauge was moved at a certain speed ($\sim 10 \text{ mm/s}$) parallel to coating, where the shear force was recorded at the same time. The same operator conducted all the measurements so that the standard error could be minimized.

Marine Field Test. The epoxy resin panels with the PDMS based coatings were immersed in seawater in inner Xiamen Bay (24 $^{\circ}$ 33'N, 118 $^{\circ}$ 09'E, China). The immersion depth was 1 m. A panel without any coating was set as control. According to ASTM D6990-05 (2011),

these panels were taken out of seawater, and the adhered fouling organisms were observed and photographed. The fouling organisms were washed by a sponge to determine whether the coating was capable of releasing them. The area covered by biofouling were determined by ImageJ.

RESULTS AND DISCUSSION

Scheme 1 shows the preparation of the PDMS-PUa/FND coating. ND with hydroxyl on the surface was modified as FND through the condensation of fluorosilane. Subsequently, FND was simply mixed with PDMS-PUa in THF under ultrasonication, and the mixture was applied on the substrate to prepare the coating.

The nanoparticles ND and FND were characterized. Figure 1a shows their Raman spectra. The peak at 1324 cm^{-1} in the spectra of ND is attributed to a standard diamond crystal with sp^3 -coordinated carbon atoms.³⁸ The peak at 1603 cm^{-1} is assigned to crystalline graphite, which is also an allotrope of carbon. The graphite is generated during the preparation of nanodiamond by the explosive method due to the high activation energy for solid state transformations.³⁹ The spectra of FND are nearly the same as those of ND, indicating that FND retains its diamond structure after modification. This is consistent with reported result.³⁸ From the TEM and SEM images (Figure 1b and Figure S2), it can be observed that ND has a particle size mostly ranging from 30 to 100 nm. FND has a similar particle size to ND because the fluoro chains on the surface are short (molar mass: $\sim 600\text{ g mol}^{-1}$), so their influence on particle size is slight. ND and FND with small sizes have large specific surface area, which is beneficial for their sufficient contact with polymer matrix and thus the reinforcement.⁴⁰

Figure 1c shows the FTIR spectra. In the spectra of ND, the broad peak observed at 3450 cm^{-1} is assigned to $-\text{OH}$, and the peaks at 2964 and 2856 cm^{-1} are attributed to $\text{C}-\text{H}$. For FND, new peaks occur at 1151 and 1211 cm^{-1} , which are assigned to the $\text{Si}-\text{O}-\text{C}$ bond and $\text{C}-\text{F}$, respectively. The facts indicate that the fluorosilane moieties were grafted onto the nanodiamond surface. The TGA curves of the nanoparticles are shown in Figure 1d. A weight loss of 15 wt % is observed on ND at $800\text{ }^\circ\text{C}$, attributed to the loss of water molecules, $-\text{OH}$, and $-\text{COOH}$ on the ND surface. In contrast, FND has 25 wt % weight loss at $800\text{ }^\circ\text{C}$, indicating the destruction of the fluorosilane moieties besides water and oxygen-containing groups. Additionally, the weight of FND drastically decreases from 320 to $450\text{ }^\circ\text{C}$, which is attributed to the decomposition of $\text{Si}-\text{O}$ and $\text{C}-\text{F}$ bonds. Because the fluorosilane FAS17 has a boiling point of $210\text{ }^\circ\text{C}$ and there is no weight loss shift until $320\text{ }^\circ\text{C}$, it can be proved that fluorosilane is covalently attached to the ND surface, not physically adsorbed.

Adhesion between the coating and substrate is critical to practical application in a marine environment. We studied the adhesion strength of the PDMS-based coatings on an epoxy panel (Figure 2). PDMS elastomer has a poor adhesion of 0.5 MPa due to its nonpolar structure, which makes the coating detach easily from the substrate under strong shear force.³⁰ PDMS-PUa has a significantly higher adhesion (1.5 MPa), probably because of the hydrogen bonding between urea and hydroxyl on the epoxy substrate.⁴¹ The incorporation of ND (0.05 – $0.2\text{ wt } \%$) improves the adhesion of PDMS-PUa (Figure S3), and the coating with $0.1\text{ wt } \%$ of ND (PDMS-PUa/ND) shows the highest adhesion of 2.4 MPa , increased by 60%. The

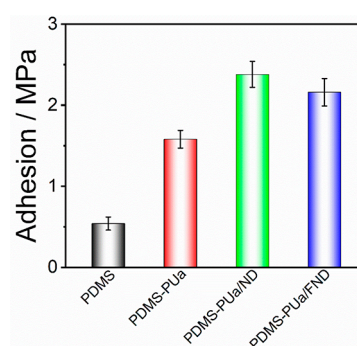


Figure 2. Adhesion strength of PDMS-based coatings on an epoxy panel.

improvement is because ND surface contains abundant hydroxyl groups which can form hydrogen bonds with the substrate and promotes the adhesion. On the other hand, nanoparticle content profoundly influences the properties of polymeric composites. At $0.1\text{ wt } \%$, ND content is enough and the aggregation is slight, so the content is optimum. Moreover, cohesive failure did not occur during the pull-off test on PDMS-PUa/ND while it happened in the case of PDMS-PUa specimens, indicating that the polyurea is reinforced. For PDMS-PUa/FND, adhesion is slightly lower than that of PDMS-PUa/ND due to the partial replacement of hydroxyl by perfluoroalkyl moieties, and fewer hydrogen bonds are formed. PDMS-PUa with ND or FND has adequate adhesion to substrate for marine applications.³⁰

Because the type of nanofiller largely affects the mechanical property of silicone coatings,³³ other nanofillers have also been tried (Figure S3). The adhesion strengths of PDMS-PUa with nanosepiolite and halloysite nanotube are 1.7 and 1.6 MPa , not so high as that of coating containing ND/FND. This indicates ND outperforms these nanofillers regarding the reinforcement of PDMS-PUa because of its high hardness. Therefore, PDMS-PUa containing $0.1\text{ wt } \%$ of ND or FND was selected for further studies including surface wettability, self-healing, and antifouling performances.

It is well-known that PDMS elastomer cannot repair itself after damage.⁴² PDMS-PUa exhibits excellent self-healing ability at room temperature due to the strong hydrogen bond among the ureas.²⁹ The influence of nanoparticles on PDMS-PUa was studied. We examined the self-healing property of PDMS-PUa/FND under ASW and in air. As shown in Figure 3a, two specimens of PDMS-PUa/FND, dyed in green and red, can merge into one piece after being cut in half and attached together for 24 h at $25\text{ }^\circ\text{C}$ under ASW. Figure 3b shows the optical microscopic picture of the surface of PDMS-PUa/FND. The cut on the surface nearly disappears after 12 h at $25\text{ }^\circ\text{C}$ and becomes invisible after 24 h . Similarly, PDMS-PUa/FND can also heal itself in air (Figure S4). Evidently, PDMS-PUa/FND is capable of self-healing.

The mechanical properties were examined by a tensile test. Figure 3c and Table 1 show the ultimate strength of the original and cut-healed samples. The incorporation of ND or FND increases their ultimate strength from 1.2 to 1.6 MPa . This is because the load is transferred from the polyurea to the nanoparticles with high hardness, so the modified PDMS-PUa can bear higher stress.^{43,44} On the other hand, the nanoparticles may serve as cross-linkers to bridge PDMS–polyurea chains, which also contributes to the improved mechanical performance. Note that PDMS-PUa can completely repair

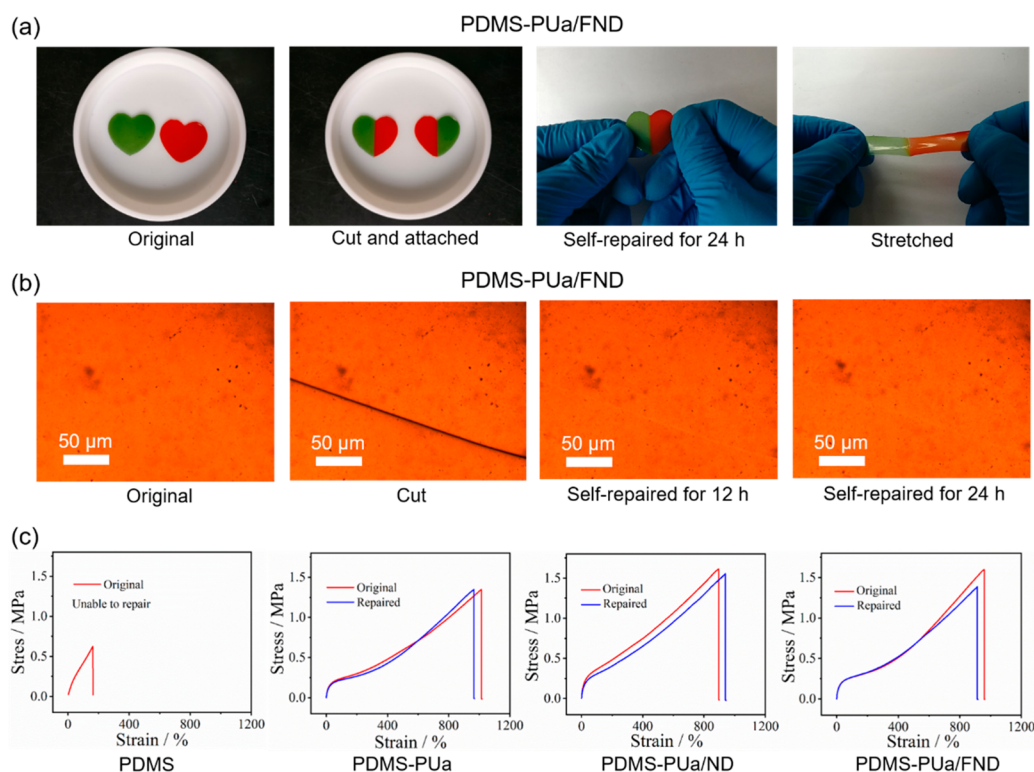


Figure 3. Cut and attached specimens of PDMS-PUa/FND (a), cut and self-repaired surfaces of PDMS-PUa/FND (b), and stress–strain curves of PDMS-based materials (c), where the healing processes were at 25 °C under ASW. The healing time for (c) was 48 h.

Table 1. Mechanical Properties of PDMS-Based Materials

sample	ultimate strength (MPa)		elongation at break (%)		elastic modulus (MPa)	
	original	repaired ^a	original	repaired ^a	original	repaired ^a
PDMS	0.65 ± 0.09		195 ± 24		0.51 ± 0.16	
PDMS-PUa	1.22 ± 0.14	1.23 ± 0.11	998 ± 29	887 ± 121	3.28 ± 0.42	3.66 ± 0.26
PDMS-PUa-ND	1.57 ± 0.22	1.43 ± 0.25	882 ± 91	845 ± 137	3.81 ± 0.27	4.32 ± 0.09
PDMS-PUa-FND	1.59 ± 0.08	1.38 ± 0.07	924 ± 68	875 ± 79	4.01 ± 0.23	3.75 ± 0.26

^aThe sample was cut, reattached, and allowed to heal at 25 °C for 48 h.

itself, while the sample containing nanoparticles had a slightly lower self-healing efficiency. This is probably because the nanoparticles reduce the contact of ureas and thus hydrogen bonds. Anyhow, PDMS-PUa with ND or FND still has self-healing ability, with >85% tensile strength recovered after 48 h at 25 °C. The elastic modulus is critical for fouling release. The PDMS elastomer has a low modulus of 0.4 MPa because of its flexible chains. PDMS-PUa exhibits a higher modulus due to the presence of hard segments. Incorporation of ND or FND further increases the modulus to ~4 MPa. Nevertheless, the modulus is low enough for the coating to have fouling release performance.

Figure 4 shows the WCA and SE of PDMS-based coatings. The PDMS elastomer exhibits a high WCA of 110° and a low SE of 22 mJ m⁻² because of its nonpolar surface. PDMS-PUa also exhibits hydrophobicity and low SE since PDMS segments can enrich on the coating surface through self-stratification.^{12,26,27} The incorporation of ND into PDMS-PUa does not notably influence the WCA and SE, indicating that ND is trapped in the bulk of the coating as the surface is covered by PDMS. In contrast, the PDMS-PUa/FND coating exhibits a significantly higher WCA (124°) and lower SE (18 mJ m⁻²). This is because the perfluoroalkyl chains drag some FND

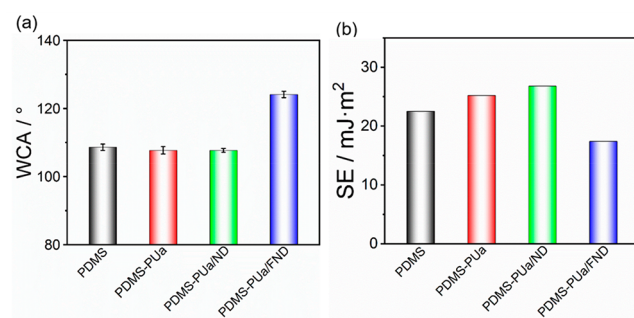


Figure 4. Water contact angle (WCA) (a) and surface energy (SE) (b) of PDMS-based coatings.

upward during coating formation and lead to a fluorinated surface. Furthermore, SEM was also applied to study whether FND is on the composite surface (Figure S5). ND cannot be seen on PDMS-PUa/ND surface, while some FND can be observed on the PDMS-PUa/FND surface. This result is in accordance with the surface wettability, showing that ND particles are buried in the bulk of composite, but some FND particles are present on surface because of the perfluoroalkyl chains. Note that Si–O–C bonds are not very stable in some

cases such as a strong basic environment. In this system, the Si—O—C bonds on FND are linked with heptadecafluoro chain (17 F atoms) with a large steric hindrance, which is beneficial for avoiding the hydrolysis.⁴⁵ Moreover, both the fluoro groups and the matrix PDMS-PUa are highly hydrophobic, so it is unlikely for the fluorosilane to come off in seawater.

We studied the surface topography and roughness of the PDMS-based coatings (Figure 5). The PDMS surface is even

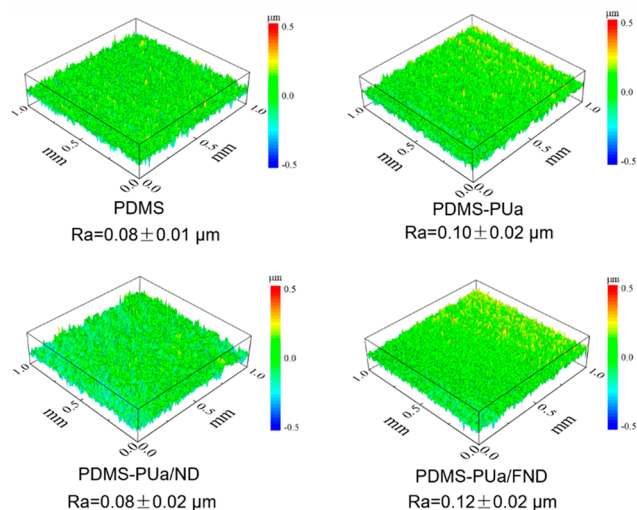


Figure 5. Surface topography and roughness of PDMS-based coatings.

and smooth with a low average roughness (R_a) of $0.08 \mu\text{m}$, which is attributed to its uniform configuration of polysiloxane chains. PDMS-PUa and PDMS-PUa/ND have very similar surface topography and R_a since their surfaces are also enriched with PDMS. For PDMS-PUa/FND, the surface is also smooth with a low R_a of $0.12 \mu\text{m}$. Although some FND particles are distributed on the coating surface as demonstrated by the WCA test, its roughness does not increase notably. This is because the FND content is very low, and the long perfluoroalkyl chains eliminate the effect of the nanoparticle on roughness. The difference of roughness among the coatings is minimal, so its influence on fouling-release ability is negligible.

Figure 6 shows the removal strength of adhered pseudobarnacles from the coating surface, which reflects the fouling-release performance. Compared with a living barnacle, the use of a pseudobarnacle reduces measurement error and enables precise control of the attachment area.⁴⁶ The removal

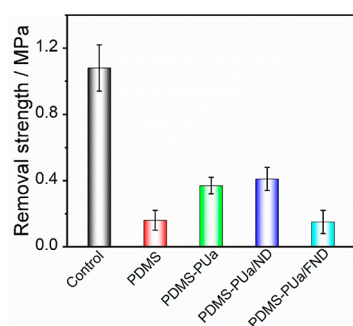


Figure 6. Removal strength of adhered pseudobarnacles from PDMS-based coatings.

strength of the pseudobarnacle from the epoxy panel control is 1.1 MPa , indicating a difficult removal. The PDMS elastomer with a low SE and low elastic modulus exhibits remarkable fouling-release ability with a removal strength lower than 0.2 MPa . PDMS-PUa and PDMS-PUa/ND have a low surface energy close to that of PDMS, but they show poorer fouling-release performance. This is because they have a higher modulus ($\sim 4 \text{ MPa}$), as shown in Table 1. PDMS-PUa/FND has a lower removal strength than PDMS. This is because the former has a lower SE (18 mJ m^{-2}) though its modulus is somewhat higher. Clearly, introduction of FND can significantly improve the fouling-release ability of PDMS-PUa.

Figure 7 shows the marine field test results of the PDMS-based coatings after immersion in Xiamen Bay, China, for four

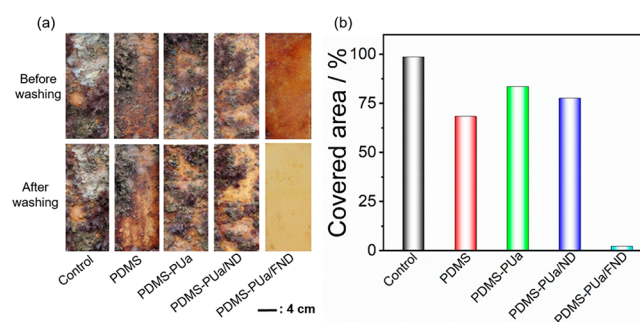


Figure 7. Images of PDMS-based coatings after immersion in natural seawater for four months (a) and the corresponding area covered by marine organisms after washing (b).

months. The fouling-release performance was evaluated by cleaning the coating surface with a sponge. The surface of the epoxy panel control after immersion was fully colonized by a variety of marine organisms, including barnacles, tubeworms, bryozoan, and algae, indicating severe fouling pressure. The PDMS elastomer was also seriously fouled, and only a few bryozoan and algae were removed, although it showed excellent efficacy in the indoor pseudobarnacle removal test. PDMS-PUa and PDMS-PUa/ND did not show good fouling-release ability either. In contrast, fewer organisms grew on PDMS-PUa/FND, probably due to the low SE. Actually, only 3% of the area was covered after cleaning. Thus, the coating exhibits good fouling release performance. The perfluoroalkyl-modified nanodiamonds in the coating greatly enhance marine antibiofouling ability.

CONCLUSION

We have reinforced poly(dimethylsiloxane)-based polyurea (PDMS-PUa) by introducing nanodiamond with chemically attached perfluoroalkyl on the surface. The modified PDMS-PUa coating is hydrophobic and has low surface energy as well as low surface roughness. The introduction of the modified nanodiamond can significantly improve the tensile strength of PDMS-PUa and its adhesion to substrate. Meanwhile, the modified PDMS-PUa can heal itself at room temperature after damage. Pseudobarnacle removal and marine field tests show the coating has excellent fouling release because of low surface energy and relatively low elastic modulus. The coating is biocide-free and can be an eco-friendly candidate to combat marine biofouling.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsapm.0c00356>.

¹H NMR spectra of PDMS-PUa, SEM images of the nanoparticles and the composites, adhesion strength of the composite coatings, and the images of self-repaired PDMS-PUa/FND (PDF)

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Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) Leonardi, A. K.; Ober, C. K. Polymer-Based Marine Antifouling and Fouling Release Surfaces: Strategies for Synthesis and Modification. *Annu. Rev. Chem. Biomol. Eng.* **2019**, *10*, 241–264.
- (2) Xie, Q. Y.; Pan, J. S.; Ma, C. F.; Zhang, G. Z. Dynamic Surface Antifouling: Mechanism and Systems. *Soft Matter* **2019**, *15*, 1087–1107.
- (3) Kardela, J. H.; Millichamp, I. S.; Ferguson, J.; Parry, A. L.; Reynolds, K. J.; Aldred, N.; Clare, A. S. Nonfreezable Water and Polymer Swelling Control the Marine Antifouling Performance of Polymers with Limited Hydrophilic Content. *ACS Appl. Mater. Interfaces* **2019**, *11*, 29477–29489.
- (4) Brisset, H.; Briand, J. F.; Barry-Martinet, R.; Duong, T. H.; Frere, P.; Gohier, F.; Leriche, P.; Bressy, C. 96X Screen-Printed Gold Electrode Platform to Evaluate Electroactive Polymers as Marine Antifouling Coatings. *Anal. Chem.* **2018**, *90*, 4978–4981.
- (5) Hu, P.; Xie, Q. Y.; Ma, C. F.; Zhang, G. Z. Silicone-Based Fouling-Release Coatings for Marine Antifouling. *Langmuir* **2020**, *36*, 2170–2183.
- (6) Lejars, M.; Margaillan, A.; Bressy, C. Fouling Release Coatings: A Nontoxic Alternative to Biocidal Antifouling Coatings. *Chem. Rev.* **2012**, *112*, 4347–4390.
- (7) Hawkins, M. L.; Grunlan, M. A. The Protein Resistance of Silicones Prepared With a PEO-Silane Amphiphile. *J. Mater. Chem.* **2012**, *22*, 19540–19546.
- (8) Guazzelli, E.; Galli, G.; Martinelli, E.; Margaillan, A.; Bressy, C. Amphiphilic Hydrolyzable Polydimethylsiloxane-b-poly-(ethyleneglycol methacrylate-co-trialkylsilyl methacrylate) Block Copolymers for Marine Coatings. I. Synthesis, Hydrolysis and Surface Wettability. *Polymer* **2020**, *186*, 121954.
- (9) Hawkins, M. L.; Faÿ, F.; Rehel, K.; Linossier, I.; Grunlan, M. A. Bacteria and Diatom Resistance of Silicones Modified with PEO-Silane Amphiphiles. *Biofouling* **2014**, *30*, 247–258.
- (10) Zeng, H. H.; Xie, Q. Y.; Ma, C. F.; Zhang, G. Z. Silicone Elastomer with Surface-Enriched, Nonleaching Amphiphilic Side Chains for Inhibiting Marine Biofouling. *ACS Appl. Polym. Mater.* **2019**, *1*, 1689–1696.
- (11) Xie, Q. Y.; Zeng, H. H.; Peng, Q. M.; Bressy, C.; Ma, C. F.; Zhang, G. Z. Self-Stratifying Silicone Coating with Nonleaching Antifoulant for Marine Anti-Biofouling. *Adv. Mater. Interfaces* **2019**, *6*, 1900535.
- (12) Xie, Q. Y.; Ma, C. F.; Liu, C.; Ma, J. L.; Zhang, G. Z. Poly(dimethylsiloxane)-Based Polyurethane with Chemically Attached Antifoulants for Durable Marine Antibiofouling. *ACS Appl. Mater. Interfaces* **2015**, *7*, 21030–21037.
- (13) Shivapooja, P.; Yu, Q.; Orihuela, B.; Mays, R.; Rittschof, D.; Genzer, J.; Lopez, G. P. Modification of Silicone Elastomer Surfaces with Zwitterionic Polymers: Short-Term Fouling Resistance and Triggered Biofouling Release. *ACS Appl. Mater. Interfaces* **2015**, *7*, 25586–25591.
- (14) Yao, J. H.; Chen, S. S.; Ma, C. F.; Zhang, G. Z. Marine Anti-Biofouling System with Poly(ϵ -caprolactone)/clay Composite as Carrier of Organic Antifoulant. *J. Mater. Chem. B* **2014**, *2*, 5100–5106.
- (15) Xu, W. T.; Ma, C. F.; Ma, J. L.; Gan, T. S.; Zhang, G. Z. Marine Biofouling Resistance of Polyurethane with Biodegradation and Hydrolyzation. *ACS Appl. Mater. Interfaces* **2014**, *6*, 4017–4024.
- (16) Chen, W. P.; Hao, D. Z.; Guo, X. L.; Hao, W. J.; Jiang, L. Preventing Diatom Adhesion Using a Hydrogel with an Orthosilicic Acid Analog as a Deceptive Food. *J. Mater. Chem. A* **2018**, *6*, 19125–19132.
- (17) Yang, W. J.; Cai, T.; Neoh, K. G.; Kang, E. T.; Dickinson, G. H.; Teo, S. L.; Rittschof, D. Biomimetic Anchors for Antifouling and Antibacterial Polymer Brushes on Stainless Steel. *Langmuir* **2011**, *27*, 7065–7076.
- (18) Salta, M.; Wharton, J. A.; Stoodley, P.; Dennington, S. P.; Goode, L. R.; Werwinski, S.; Mart, U.; Wood, R. J.; Stokes, K. R. Designing Biomimetic Antifouling Surfaces. *Philos. Trans. R. Soc., A* **2010**, *368*, 4729–4754.
- (19) Amini, S.; Kolke, S.; Petrone, L.; Ahanotu, O.; Sunny, S.; Sutanto, C. N.; Hoon, S.; Cohen, L.; Weaver, J. C.; Aizenberg, J.; Vogel, N.; Miserez, A. Preventing Mussel Adhesion Using Lubricant-Infused Materials. *Science* **2017**, *357*, 668–673.
- (20) Shivapooja, P.; Wang, Q.; Orihuela, B.; Rittschof, D.; López, G. P.; Zhao, X. Bioinspired Surfaces with Dynamic Topography for Active Control of Biofouling. *Adv. Mater.* **2013**, *25*, 1430–1434.
- (21) Cao, X.; Pettitt, M. E.; Wode, F.; Arpa Sancet, M. P.; Fu, J.; Ji, J.; Callow, M. E.; Callow, J. A.; Rosenhahn, A.; Grunze, M. Interaction of Zoospores of the Green Alga Ulva with Bioinspired Micro- and Nanostructured Surfaces Prepared by Polyelectrolyte Layer-by-Layer Self-Assembly. *Adv. Funct. Mater.* **2010**, *20*, 1984–1993.
- (22) Qian, P. Y.; Li, Z.; Xu, Y.; Li, Y.; Fusetani, N. Mini-Review: Marine Natural Products and Their Synthetic Analogs as Antifouling Compounds: 2009–2014. *Biofouling* **2015**, *31*, 101–122.

- (23) Hunsucker, K. Z.; Braga, C.; Gardner, H.; Jongerius, M.; Hietbrink, R.; Salters, B.; Swain, G. Using Ultraviolet Light for Improved Antifouling Performance on Ship Hull Coatings. *Biofouling* **2019**, *35*, 658–668.
- (24) Guo, S. F.; Lee, H. P.; Chaw, K. C.; Miklas, J.; Teo, S. L. M.; Dickinson, G. H.; Birch, W. R.; Khoo, B. C. Effect of Ultrasound on Cyprids and Juvenile Barnacles. *Biofouling* **2011**, *27*, 185–192.
- (25) Park, J. S.; Lee, J. H. Sea-Trial Verification of Ultrasonic Antifouling Control. *Biofouling* **2018**, *34*, 98–110.
- (26) Majumdar, P.; Stafslin, S.; Daniels, J.; Webster, D. C. High Throughput Combinatorial Characterization of Thermosetting Siloxane–Urethane Coatings Having Spontaneously Formed Microtopographical Surfaces. *J. Coat. Technol. Res.* **2007**, *4*, 131–138.
- (27) Fang, J.; Kelarakis, A.; Wang, D. Y.; Giannelis, E. P.; Finlay, J. A.; Callow, M. E.; Callow, J. A. Fouling Release Nanostructured Coatings Based on PDMS-polyurea Segmented Copolymers. *Polymer* **2010**, *51*, 2636–2642.
- (28) Chen, Z. G.; Chisholm, B.; Kim, J.; Stafslin, S.; Wagner, R.; Patel, S.; Daniels, J.; Wal, L. V.; Li, J.; Ward, K.; Callow, M.; Thompson, S.; Siripirom, C. UV-Curable, Oxetane-Toughened Epoxy-Siloxane Coatings for Marine Fouling-Release Coating Applications. *Polym. Int.* **2008**, *57*, 879–886.
- (29) Liu, C.; Ma, C. F.; Xie, Q. Y.; Zhang, G. Z. Self-Repairing Silicone Coatings for Marine Anti-Biofouling. *J. Mater. Chem. A* **2017**, *5*, 15855–15861.
- (30) Liu, C.; Xie, Q. Y.; Ma, C. F.; Zhang, G. Z. Fouling Release Property of Polydimethylsiloxane-Based Polyurea with Improved Adhesion to Substrate. *Ind. Eng. Chem. Res.* **2016**, *55*, 6671–6676.
- (31) Cavas, L.; Yildiz, P. G.; Mimigianni, P.; Sapalidis, A.; Nitodas, S. Reinforcement Effects of Multiwall Carbon Nanotubes and Graphene Oxide on PDMS Marine Coatings. *J. Coat. Technol. Res.* **2018**, *15*, 105–120.
- (32) Selim, M. S.; El-Safty, S. A.; Fatthallah, N. A.; Shenashen, M. A. Silicone/Graphene Oxide Sheet-Alumina Nanorod Ternary Composite for Superhydrophobic Antifouling Coating. *Prog. Org. Coat.* **2018**, *121*, 160–172.
- (33) Selim, M. S.; Selim, M. S.; Shenashen, M. A.; El-Safty, S. A.; Higazy, S. A.; Selim, M. M.; Isago, H.; Elmarakbi, A. Recent Progress in Marine Foul-Release Polymeric Nanocomposite Coatings. *Prog. Mater. Sci.* **2017**, *87*, 1–32.
- (34) Behler, K. D.; Stravato, A.; Mochalin, V.; Korneva, G.; Yushin, G.; Gogotsi, Y. Nanodiamond-Polymer Composite Fibers and Coatings. *ACS Nano* **2009**, *3*, 363–369.
- (35) Zheng, W. W.; Hsieh, Y. H.; Chiu, Y. C.; Cai, S. J.; Cheng, C. L.; Chen, C. Organic Functionalization of Ultradispersed Nanodiamond: Synthesis and Applications. *J. Mater. Chem.* **2009**, *19*, 8432–8441.
- (36) Liang, Y. J.; Ozawa, M.; Krueger, A. A General Procedure to Functionalize Agglomerating Nanoparticles Demonstrated on Nanodiamond. *ACS Nano* **2009**, *3*, 2288–2296.
- (37) Bokobza, L.; Rahmani, M. Carbon Nanotubes: Exceptional Reinforcing Fillers for Silicone Rubbers. *Kautsch. Gummi. Kunstst.* **2009**, *62*, 112–117.
- (38) Panich, A. M.; Vieth, H. M.; Shames, A. I.; Froumin, N.; Osawa, E.; Yao, A. Structure and Bonding in Fluorinated Nanodiamond. *J. Phys. Chem. C* **2015**, *114*, 774–782.
- (39) Shenderova, O. A.; Gruen, D. M. *Ultranacocrystalline Diamond: Synthesis, Properties and Application*; William Andrew: Norwich, 2006.
- (40) Balazs, A. C.; Emrick, T.; Russell, T. P. Nanoparticle Polymer Composites: Where Two Small Worlds Meet. *Science* **2006**, *314*, 1107–1110.
- (41) Awaja, F.; Gilbert, M.; Kelly, G.; Fox, B.; Pigram, P. J. Adhesion of Polymers. *Prog. Polym. Sci.* **2009**, *34*, 948–968.
- (42) Keller, M. W.; White, S. R.; Sottos, N. R. A Self-healing Poly(dimethyl siloxane) Elastomer. *Adv. Funct. Mater.* **2007**, *17*, 2399–2404.
- (43) Cadek, M.; Coleman, J. N.; Barron, V.; Hedicke, K.; Blau, W. J. Morphological and Mechanical Properties of Carbon-nanotube-reinforced Semicrystalline and Amorphous Polymer Composites. *Appl. Phys. Lett.* **2002**, *81*, 5123–5125.
- (44) Vivekchand, S. R. C.; Ramamurty, U.; Rao, C. N. R. Mechanical Properties of Inorganic Nanowire Reinforced Polymer–Matrix Composites. *Nanotechnology* **2006**, *17*, 344–350.
- (45) Li, Y. J.; Zhang, C. C.; Zhou, Y.; Chen, Y. J.; Dong, Y. J. Multiresponsive Aggregates Based on a Sensitive Si–O–C Structure: When the Chemical Bond Nature Meets Self-assembly. *Chem. Lett.* **2016**, *45*, 904–906.
- (46) Stein, J.; Truby, K.; Wood, C. D.; Takemori, M.; Vallance, M.; Swain, G.; Kavanagh, C.; Kovach, B.; Schultz, M.; Wiebe, D.; Holm, E.; Montemarano, J.; Wendt, D.; Smith, C.; Meyer, A. Structure-Property Relationships of Silicone Biofouling-Release Coatings: Effect of Silicone Network Architecture on Pseudobarnacle Attachment Strengths. *Biofouling* **2003**, *19*, 87–94.